

Lab 4 and 5 and 6 - Rishvanth SK, 240958138 Roll: 35

Aim

To design and implement RESTful web services using Spring Boot and integrate them with a MySQL database to perform CRUD operations on multiple customer-related modules.

Objective

- To understand REST API design principles.
- To implement CRUD operations using HTTP methods (POST, GET, PUT).
- To apply layered architecture in backend development.
- To integrate Spring Boot with MySQL using JPA and Hibernate.
- To analyse application startup flow and database connectivity behaviour.

Theoretical Background

REST API Design

- Resources are identified using URIs.
- HTTP methods define operations on resources.
- Communication is stateless.
- Responses include meaningful status codes.

Example

- GET /customers – Retrieve all customers
- POST /customers – Create new customer
- PUT /customers/{id} – Update customer
- DELETE /customers/{id} – Delete customer

Layered Architecture

All experiments followed a structured layered pattern:

- Controller Layer – Handles HTTP requests.
- Service Layer – Contains business logic.
- Repository Layer – Interacts with the database via JPA.
- Database Layer – MySQL storage.

Request Flow: Client → Controller → Service → Repository → Database → Response

Spring Boot uses auto-configuration and DispatcherServlet to manage request routing.

Work Done (Lab-wise Progression)

Lab 4 – REST API Implementation

- Designed REST endpoints for multiple customer-related modules.
- Implemented HTTP methods (GET, POST, PUT).

- Tested APIs using Postman.
- Verified correct status codes and responses.
- This lab focused on understanding request-response flow and stateless communication.

Lab 5 – CRUD Application Development

A complete CRUD-based backend was developed for the following modules:

1. CustomerDetail
2. CustomerIdentification
3. CustomerContactInformation
4. CustomerProofOfId
5. CustomerAddress Each module implemented:

- POST – Create record
 - GET – Retrieve all / by ID
 - PUT – Update record
- The system was structured using: Entity → DTO → Repository → Service → Controller
This ensured separation of concerns and modular design\

Lab 6 – Database Integration & Testing

Configured Spring Boot to connect with MySQL using application properties.

Hibernate was used as ORM for table mapping.

Automatic table generation was enabled.

API endpoints were tested for database persistence. During execution, a database connectivity issue was encountered (missing database).

The issue was diagnosed through stack trace analysis and resolved by creating the required database in MySQL. After correction:

- Application started successfully.
- Tables were created.
- CRUD operations functioned correctly.

5. Learning Outcomes

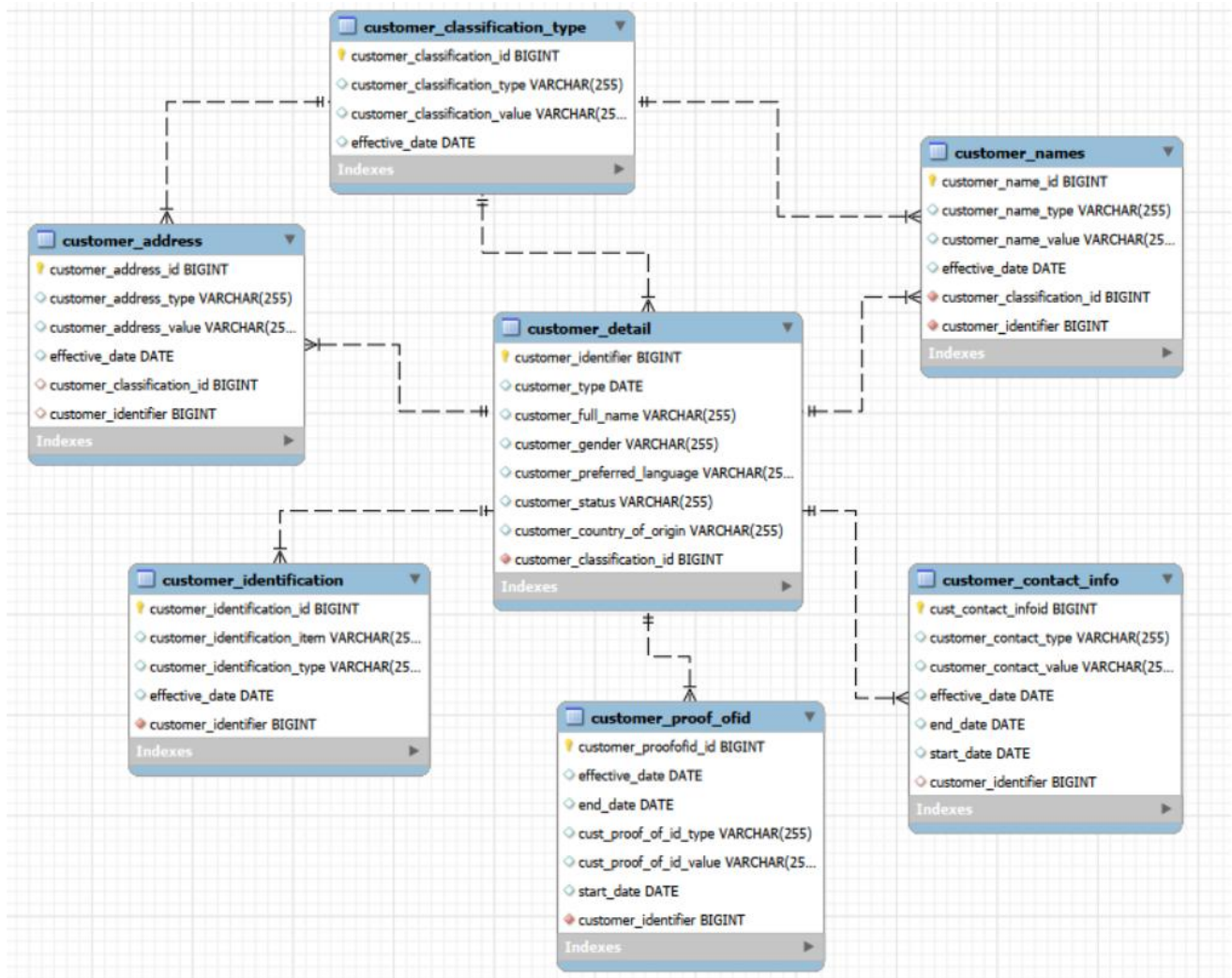
- Understanding of REST API design principles.
- Implementation of CRUD operations using Spring Boot.
- Practical use of layered architecture.
- Integration of Spring Boot with MySQL using JPA.
- Importance of correct database configuration.
- Debugging startup failures using stack trace analysis.
- Understanding how Hibernate determines SQL dialect.

6. Conclusion

Labs 4, 5, and 6 provided a comprehensive understanding of backend application development using Spring Boot.

The progression from REST API design (Lab 4), to full CRUD implementation (Lab 5), and finally database integration and troubleshooting (Lab 6), established a complete learning cycle.

These labs demonstrated how scalable and maintainable backend systems are structured in real-world enterprise applications using REST principles and layered architecture.



Lab4and5and6 [boot]

src/main/java

- com.example.demo
 - Lab4and5and6Application.java
- com.example.demo.controllers
 - CustomerAddressController.java
 - CustomerClassificationController.java
 - CustomerContactInfoController.java
 - CustomerDetailController.java
 - CustomerIdentificationController.java
 - CustomerNameController.java
 - CustomerProofOfIdController.java
- com.example.demo.DTO
 - custAddressDTO.java
 - custClassifDTO.java
 - custContactInfoDTO.java
 - custDetailDTO.java
 - custIdentifDTO.java
 - custNameDTO.java
 - custProofOfIdDTO.java

com.example.demo.entity

- CustomerAddress.java
- CustomerClassificationType.java
- CustomerContactInformation.java
- CustomerDetail.java
- CustomerIdentification.java
- CustomerName.java
- CustomerProofOfId.java
- com.example.demo.mapping
 - CustDetailMapper.java
- com.example.demo.repository
 - CustomerClassificationRepo.java
 - CustomerContactInformationRepo.java
 - CustomerDetailRepo.java
 - CustomerIdentificationRepo.java
 - CustomerNameRepo.java
 - CustomerProofOfIdRepo.java
 - CutomerAddressRepo.java
- com.example.demo.service
 - CustomerAddressService.java
 - CustomerClassificationService.java
 - CustomerContactInfoService.java
 - CustomerDetailsService.java
 - CustomerIdentificationService.java
 - CustomerNameService.java

com.example.demo.service

- CustomerAddressService.java
- CustomerClassificationService.java
- CustomerContactInfoService.java
- CustomerDetailsService.java
- CustomerIdentificationService.java
- CustomerNameService.java
- CustomerProofOfIdService.java
- com.example.demo.serviceImpl
 - custDetailsServiceImpl.java
 - CustomerAddressServiceImpl.java
 - CustomerClassificationServiceImpl.java
 - CustomerContactInfoServiceImpl.java
 - CustomerIdentificationServiceImpl.java
 - CustomerNameServiceImpl.java
 - CustomerProofOfIdServiceImpl.java

Starting with **customer classification**

POST:

POST

http://localhost:8080/api/classification

Send

Docs

Params

Auth

Headers (8)

Body

Scripts

Settings

Cookies

raw

JSON

Schema

Beautify

```
1 {
2   "customerClassificationType": "RETAIL",
3   "customerClassificationValue": "INDIVIDUAL",
4   "effectiveDate": "2024-01-01"
5 }
6
```

Body

200 OK

350 ms

373 B

Save Response

JSON

Preview

Visualize

```
1 {
2   "customerClassificationId": 1,
3   "customerClassificationType": "RETAIL",
4   "customerClassificationValue": "INDIVIDUAL",
5   "effectiveDate": "2024-01-01",
```

Similarly generated in the table:

```
mysql> select * from customer_classification_type;
+-----+-----+-----+-----+-----+
| customer_classification_id | crud_value | customer_classification_type | customer_classification_value | effective_date |
+-----+-----+-----+-----+-----+
| 1 | C | RETAIL | INDIVIDUAL | 2024-01-01 |
+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)
```

GET:

GET

http://localhost:8080/api/classification

Send

Docs

Params

Auth

Headers (6)

Body

Scripts

Settings

Cookies

Query Params

Key	Value	Description
Key	Value	Description

Body

200 OK

1.75 s

793 B

Save Response

JSON

Preview

Visualize

```
1 [
2   {
3     "customerClassificationId": 1,
4     "customerClassificationType": "RETAIL",
5     "customerClassificationValue": "INDIVIDUAL",
6     "effectiveDate": "2024-01-01",
7     "customerDetails": [],
8     "customerNames": null,
9     "customerAddresses": null
10  },
11  {
12    "customerClassificationId": 2,
13    "customerClassificationType": "NON_RETAIL",
14    "customerClassificationValue": "CORPORATE",
15    "effectiveDate": "2024-02-15",
16    "customerDetails": [],
17    "customerNames": null,
18    "customerAddresses": null
19  },
20 ]
```

Some attributes are null because they haven't been created yet. Once done they will be linked

PUT:

PUT

http://localhost:8080/api/classification/1

Send

DocsParamsAuthHeaders (8)BodyScriptsSettings

Cookies

rawJSON

SchemaBeautify

1{

2 "customerClassificationType": "RETAIL",

3 "customerClassificationValue": "MSME",

4 "effectiveDate": "2024-07-20"

5}

Body

200 OK849 ms367 B

Save Response

{JSON

PreviewVisualize

1{

2 "customerClassificationId": 1,

3 "customerClassificationType": "RETAIL",

4 "customerClassificationValue": "MSME",

5 "effectiveDate": "2024-07-20",

6 "customerDetails": null,

7 "customerNames": null,

8 "customerAddresses": null

9}

mysql> select * from customer_classification_type;

customer_classification_id	crud_value	customer_classification_type	customer_classification_value	effective_date
1	U	RETAIL	MSME	2024-07-20
2	C	NON_RETAIL	CORPORATE	2024-02-15
3	C	INSTITUTIONAL	BANK	2024-01-10

3 rows in set (0.01 sec)

On updating the CRUD value is changed from C-create to U-updated

DELETE:

DELETE

http://localhost:8080/api/classification/2

Send

DocsParamsAuthHeaders (6)BodyScriptsSettings

Cookies

Body

200 OK776 ms184 B

Save Response

Raw

PreviewVisualize

1Deleted successfully

Trying to delete the same id again will throw an exception

Body

400 Bad Request

Raw

PreviewDebug with AI

1Classification already deleted

The CRUD value in the database changes:

mysql> select * from customer_classification_type;

customer_classification_id	crud_value	customer_classification_type	customer_classification_value	effective_date
1	U	RETAIL	MSME	2024-07-20
2	D	NON_RETAIL	CORPORATE	2024-02-15
3	C	INSTITUTIONAL	BANK	2024-01-10

3 rows in set (0.01 sec)

This deleted entity wont be visible in get request also:

[

{

"customerClassificationId": 1,

"customerClassificationType": "RETAIL",

"customerClassificationValue": "MSME",

"effectiveDate": "2024-07-20",

"customerDetails": [],

"customerNames": null,

"customerAddresses": null

,

{

"customerClassificationId": 3,

"customerClassificationType": "INSTITUTIONAL",

"customerClassificationValue": "BANK",

"effectiveDate": "2024-01-10",

"customerDetails": [],

"customerNames": null,

"customerAddresses": null

]

Customer detail: Main class which will interact with the user and store all the details

POST request:

```
{
  "custFullName": "Priya Sharma",
  "custGender": "FEMALE",
  "custDate": "1995-03-12",
  "custPrefLanguage": "HI",
  "custStatus": "ACTIVE",
  "custCountry": "India",
  "classificationId": 1,

  "customerAddresses": [
    {
      "addressType": "HOME",
      "addressValue": "Andheri East, Mumbai, Maharashtra",
      "effectiveDate": "2024-02-01"
    }
  ],

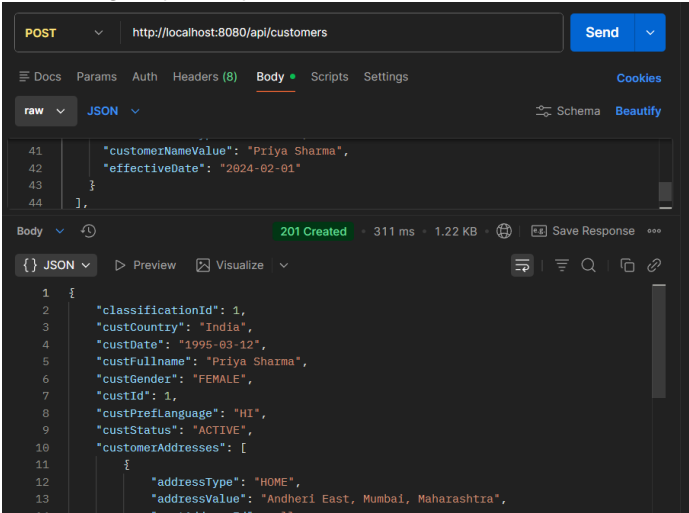
  "customerContactInformations": [
    {
      "customerContactType": "PHONE",
      "customerContactValue": "+919876543210",
      "effectiveDate": "2024-02-01",
      "startDate": "2024-02-01",
      "endDate": "2027-02-01"
    }
  ],

  "customerProofOfIds": [
    {
      "proofOfIdType": "PASSPORT",
      "proofOfIdValue": "M1234567",
      "effectiveDate": "2024-02-01",
      "startDate": "2024-02-01",
      "endDate": "2034-02-01"
    }
  ]
},

{
  "customerNames": [
    {
      "customerNameType": "PRIMARY",
      "customerNameValue": "Priya Sharma",
      "effectiveDate": "2024-02-01"
    }
  ],

  "identification": {
    "custIdentificationtype": "VOTER_ID",
    "custIdentificationItem": "MH1234567890",
    "effectiveDate": "2024-02-01"
  }
}
```

On sending the post request:



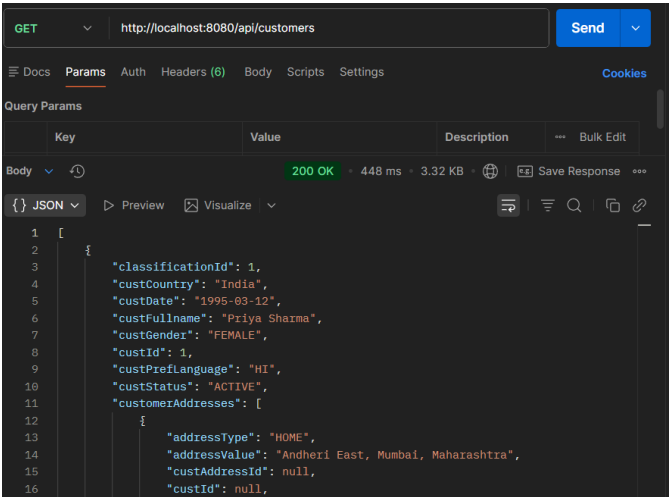
Similarly created in the database

```
mysql> select * from customer_detail;
```

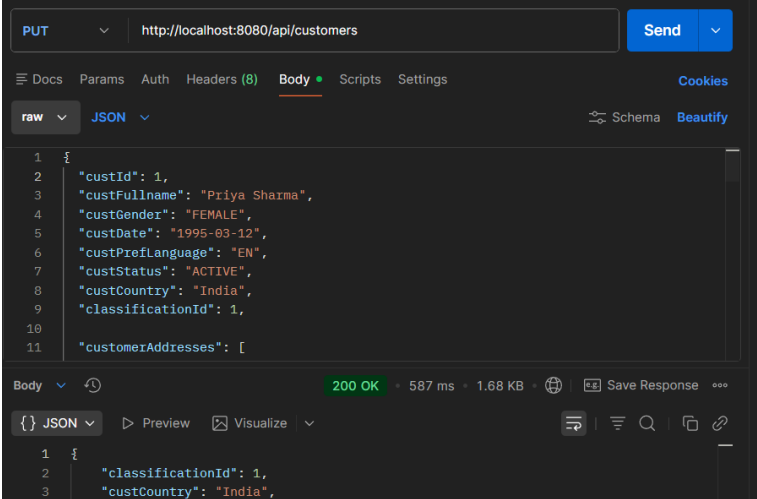
customer_identifier	crud_value	customer_type	customer_full_name	customer_gender	customer_preferred_language	customer_status	customer_country_of_origin	customer_classification_id
1	C	1995-03-12	Priya Sharma	FEMALE	HI	ACTIVE	India	1

1 row in set (0.04 sec)

GET:



PUT:



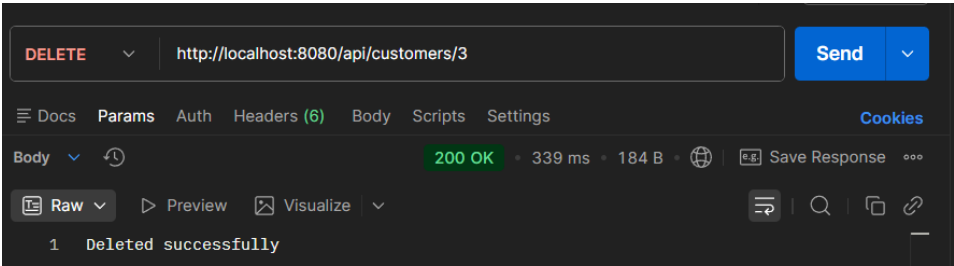
On updating the details the CRUD value is changed in the database

```
mysql> select * from customer_detail;
```

customer_identifier	crud_value	customer_type	customer_full_name	customer_gender	customer_preferred_language	customer_status	customer_country_of_origin	customer_classification_id
1	U	1995-03-12	Priya Sharma	FEMALE	EN	ACTIVE	India	1
2	C	1990-11-22	Rahul Menon	MALE	EN	ACTIVE	UAE	3
3	C	1997-07-18	Sanjay Iyer	MALE	TA	ACTIVE	India	3

3 rows in set (0.01 sec)

DELETE:



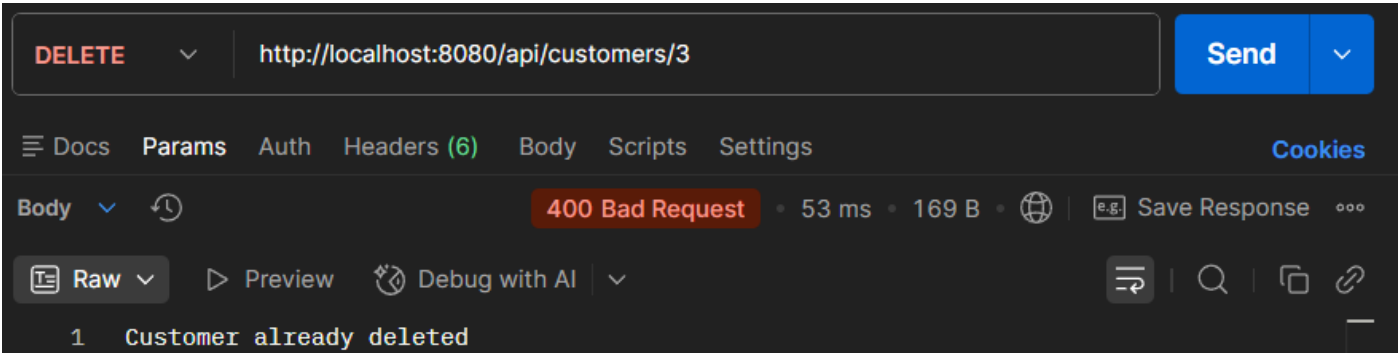
```
mysql> select * from customer_detail;
```

customer_identifier	crud_value	customer_type	customer_full_name	customer_gender	customer_preferred_language	customer_status	customer_country_of_origin	customer_classification_id
1	U	1995-03-12	Priya Sharma	FEMALE	EN	ACTIVE	India	1
2	C	1990-11-22	Rahul Menon	MALE	EN	ACTIVE	UAE	3
3	D	1997-07-18	Sanjay Iyer	MALE	TA	ACTIVE	India	3

3 rows in set (0.01 sec)

CRUD value is changed

Cannot send the same request again



When the customer detail is created, even the other tables are populated with the corresponding details

```
mysql> select * from customer_identification;
```

customer_identification_id	crud_value	customer_identification_item	customer_identification_type	effective_date	customer_identifier
1	U	MH1234567890	VOTER_ID	2025-01-01	1
2	C	Z9876543	PASSPORT	2024-04-01	2
3	D	1234-5678-9999	AADHAAR	2024-01-01	3

3 rows in set (0.04 sec)

```
mysql> select * from customer_address;
```

customer_address_id	customer_address_type	customer_address_value	crud_value	effective_date	customer_classification_id	customer_identifier
2	CURRENT	Dubai Marina, Dubai, UAE	C	2024-04-01	3	2
3	HOME	T Nagar, Chennai, Tamil Nadu	D	2024-01-01	2	3
4	HOME	Powai, Mumbai, Maharashtra	C	2025-01-01	1	1
5	OFFICE	BKC, Mumbai, Maharashtra	C	2025-01-01	1	1

4 rows in set (0.01 sec)

```
mysql> select * from customer_contact_info;
```

cust_contact_info_id	crud_value	customer_contact_type	customer_contact_value	effective_date	end_date	start_date	customer_identifier
2	C	EMAIL	rahul.menon@example.com	2024-04-01	2028-04-01	2024-04-01	2
3	D	EMAIL	sanjay.iyer@example.com	2024-01-01	2026-01-01	2024-01-01	3
4	C	PHONE	+919812345678	2025-01-01	2028-01-01	2025-01-01	1
5	C	EMAIL	priya.sharma@newmail.com	2025-01-01	2028-01-01	2025-01-01	1

```
4 rows in set (0.02 sec)
```

```
mysql> select * from customer_proof_ofid;
```

customer_proofofid_id	crud_value	effective_date	end_date	cust_proof_of_id_type	cust_proof_of_id_value	start_date	customer_identifier
2	C	2024-04-01	2034-04-01	PASSPORT	Z9876543	2024-04-01	2
3	D	2024-01-01	2034-01-01	PAN	SANPI1234K	2024-01-01	3
4	C	NULL	2035-01-01	PASSPORT	M1234567	2025-01-01	1

```
3 rows in set (0.01 sec)
```

```
mysql> select * from customer_names;
```

customer_name_id	crud_value	customer_name_type	customer_name_value	effective_date	customer_classification_id	customer_identifier
2	C	PRIMARY	Rahul Menon	2024-04-01	3	2
3	D	PRIMARY	Sanjay Iyer	2024-01-01	2	3
4	C	PRIMARY	Priya Sharma	2025-01-01	1	1
5	C	ALIAS	Priya S. Sharma	2025-01-01	1	1

```
4 rows in set (0.02 sec)
```

On top of this we can even perform the crud operations on all the other entity classes separately

Customer Address:

POST:

```
{
  "custId": 1,
  "customerClassificationId": 1,
  "addressType": "HOME",
  "addressValue": "Koramangala, Bengaluru, Karnataka",
  "effectiveDate": "2025-01-01"
}
```

POST `http://localhost:8080/api/address` **Send**

Docs Params Auth Headers (8) **Body** Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "custId": 1,
3   "customerClassificationId": 1,
4   "addressType": "HOME",
5   "addressValue": "Koramangala, Bengaluru, Karnataka",
6   "effectiveDate": "2025-01-01"
7 }
```

Body **201 Created** • 153 ms • 329 B • Save Response

{ } JSON Preview Visualize

```
1 {
2   "addressType": "HOME",
3   "addressValue": "Koramangala, Bengaluru, Karnataka",
4   "custAddressId": 6,
5   "custId": 1,
6   "customerClassificationId": 1,
7   "effectiveDate": "2025-01-01"
8 }
```

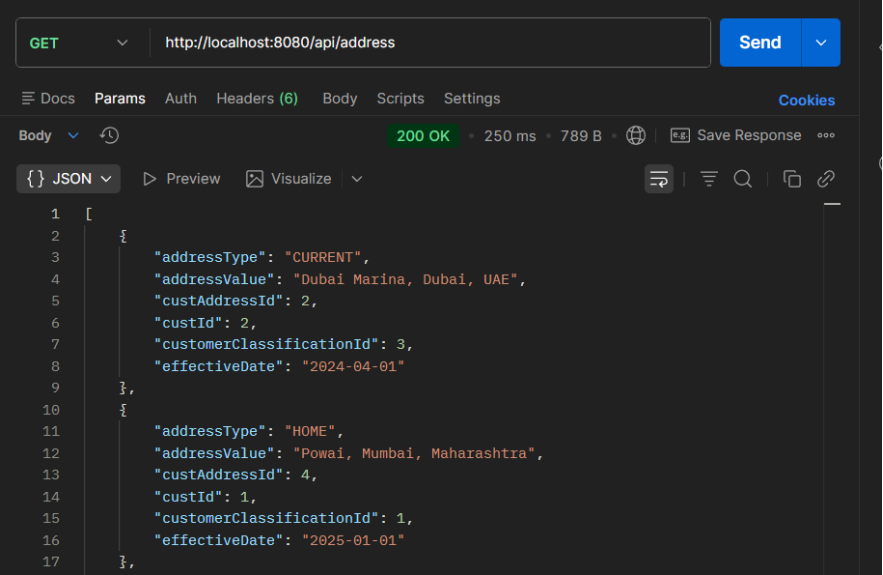
Created in the table

```
mysql> select * from customer_address;
```

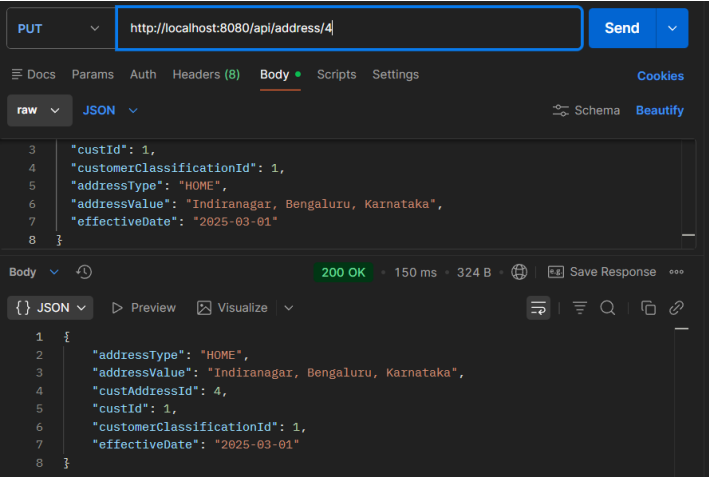
customer_address_id	customer_address_type	customer_address_value	crud_value	effective_date	customer_classification_id	customer_identifier
2	CURRENT	Dubai Marina, Dubai, UAE	C	2024-04-01	3	2
3	HOME	T Nagar, Chennai, Tamil Nadu	D	2024-01-01	2	3
4	HOME	Powai, Mumbai, Maharashtra	C	2025-01-01	1	1
5	OFFICE	BKC, Mumbai, Maharashtra	C	2025-01-01	1	1
6	HOME	Koramangala, Bengaluru, Karnataka	C	2025-01-01	1	1

```
5 rows in set (0.01 sec)
```


GET:



PUT:



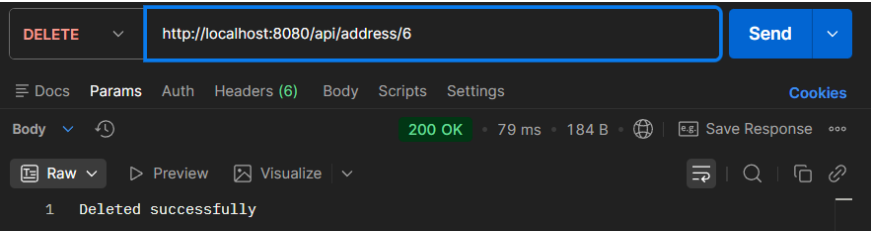
Updated in the db with the CRUD value also

```
mysql> select * from customer_address;
```

customer_address_id	customer_address_type	customer_address_value	crud_value	effective_date	customer_classification_id	customer_identifier
2	CURRENT	Dubai Marina, Dubai, UAE	C	2024-04-01	3	2
3	HOME	T Nagar, Chennai, Tamil Nadu	D	2024-01-01	2	3
4	HOME	Indiranagar, Bengaluru, Karnataka	U	2025-03-01	1	1
5	OFFICE	BKC, Mumbai, Maharashtra	C	2025-01-01	1	1
6	HOME	Koramangala, Bengaluru, Karnataka	C	2025-01-01	1	1

5 rows in set (0.00 sec)

DELETE:



CRUD updated in the db:

```
mysql> select * from customer_address;
```

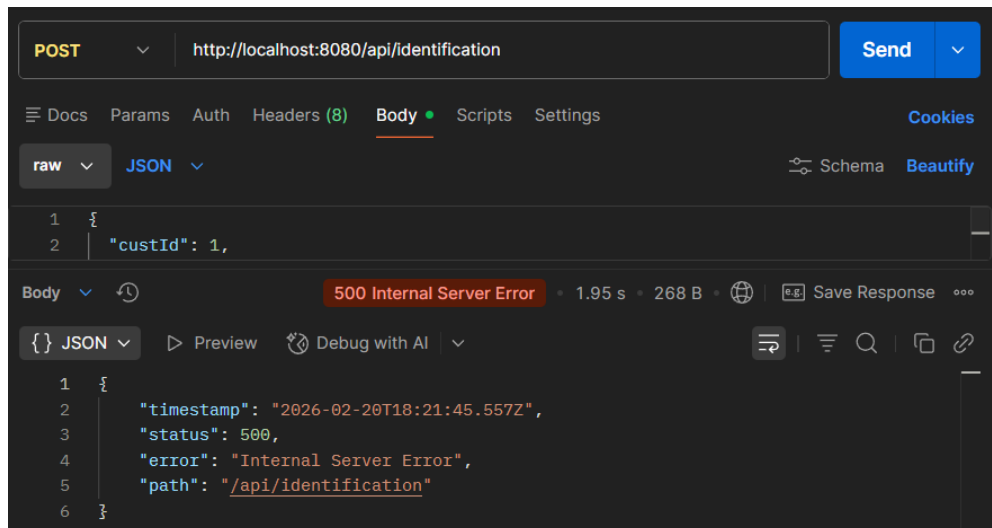
customer_address_id	customer_address_type	customer_address_value	crud_value	effective_date	customer_classification_id	customer_identifier
2	CURRENT	Dubai Marina, Dubai, UAE	C	2024-04-01	3	2
3	HOME	T Nagar, Chennai, Tamil Nadu	D	2024-01-01	2	3
4	HOME	Indiranagar, Bengaluru, Karnataka	U	2025-03-01	1	1
5	OFFICE	BKC, Mumbai, Maharashtra	C	2025-01-01	1	1
6	HOME	Koramangala, Bengaluru, Karnataka	D	2025-01-01	1	1

5 rows in set (0.01 sec)

Customer identification:

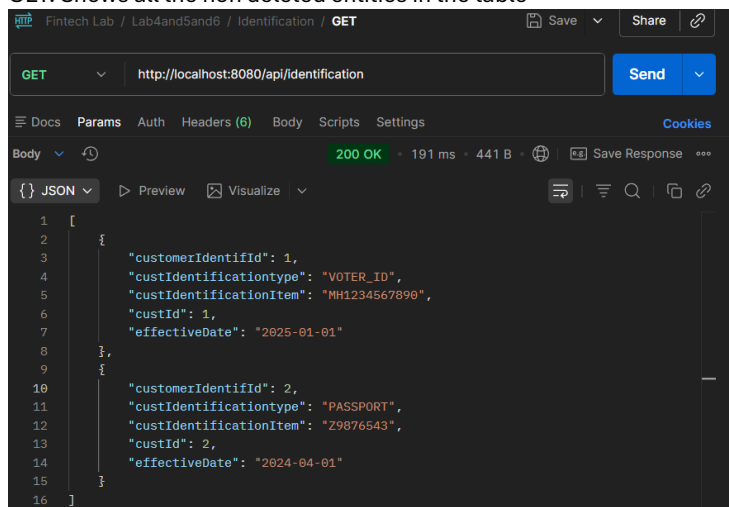
POST: because its one to one, adding one more identification entity for the same class throws an exception

```
{
  "custId": 1,
  "custIdentificationtype": "DRIVING LICENSE",
  "custIdentificationItem": "KA019999",
  "effectiveDate": "2025-02-01"
}
```

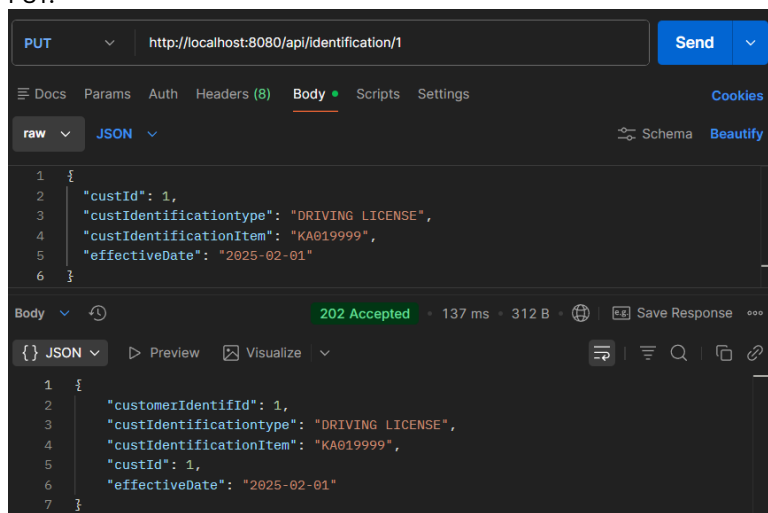


java.sql.SQLIntegrityConstraintViolationException: Duplicate entry '1' for key 'customer_identification.UKjj7jc4ie4y4xs1q8j1nj6mr9r'

GET: Shows all the non deleted entities in the table



PUT:



```
mysql> select * from customer_identification;
+-----+-----+-----+-----+-----+-----+
| customer_identification_id | crud_value | customer_identification_item | customer_identification_type | effective_date | customer_identifier |
+-----+-----+-----+-----+-----+-----+
| 1 | U | KA019999 | DRIVING LICENSE | 2025-02-01 | 1 |
| 2 | C | Z9876543 | PASSPORT | 2024-04-01 | 2 |
| 3 | D | 1234-5678-9999 | AADHAAR | 2024-01-01 | 3 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)
```

DELETE:

DELETE

http://localhost:8080/api/identification/1

Send

Docs

Params

Auth

Headers (8)

Body

Scripts

Settings

Cookies

Body

200 OK

79 ms

184 B

Save Response

Raw

Preview

Visualize

1

Successfully deleted

```
mysql> select * from customer_identification;
+-----+-----+-----+-----+-----+-----+
| customer_identification_id | crud_value | customer_identification_item | customer_identification_type | effective_date | customer_identifier |
+-----+-----+-----+-----+-----+-----+
| 1 | D | KA019999 | DRIVING LICENSE | 2025-02-01 | 1 |
| 2 | C | Z9876543 | PASSPORT | 2024-04-01 | 2 |
| 3 | D | 1234-5678-9999 | AADHAAR | 2024-01-01 | 3 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Customer name:

POST:

```
{
  "customerNameType": "PRIMARY",
  "customerNameValue": "Donald Trump",
  "effectiveDate": "2024-04-01",
  "classificationId": 3,
  "custId": 2
}
```

POST

http://localhost:8080/api/customernames

Send

Docs

Params

Auth

Headers (8)

Body

Scripts

Settings

Cookies

raw

JSON

Schema

Beautify

1

{

Body

200 OK

190 ms

305 B

Save Response

{}

JSON

Preview

Visualize

1

{

2

"custNameId": 6,

3

"customerNameType": "PRIMARY",

4

"customerNameValue": "Donald Trump",

5

"effectiveDate": "2024-04-01",

6

"classificationId": 3,

7

"custId": 2

8

}

```
mysql> select * from customer_names;
+-----+-----+-----+-----+-----+-----+
| customer_name_id | crud_value | customer_name_type | customer_name_value | effective_date | customer_classification_id | customer_identifier |
+-----+-----+-----+-----+-----+-----+
| 2 | C | PRIMARY | Rahul Menon | 2024-04-01 | 3 | 2 |
| 3 | D | PRIMARY | Sanjay Iyer | 2024-01-01 | 2 | 3 |
| 4 | C | PRIMARY | Priya Sharma | 2025-01-01 | 1 | 1 |
| 5 | C | ALIAS | Priya S. Sharma | 2025-01-01 | 1 | 1 |
| 6 | C | PRIMARY | Donald Trump | 2024-04-01 | 3 | 2 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

GET:

GET

http://localhost:8080/api/customernames

Send

Docs

Params

Auth

Headers (6)

Body

Scripts

Settings

Cookies

Body

200 OK

112 ms

733 B

Save Response

JSON

Preview

Visualize

```
1 {
2   {
3     "custNameId": 2,
4     "customerNameType": "PRIMARY",
5     "customerNameValue": "Rahul Menon",
6     "effectiveDate": "2024-04-01",
7     "classificationId": 3,
8     "custId": 2
9   },
10  {
11    "custNameId": 4,
12    "customerNameType": "PRIMARY",
13    "customerNameValue": "Priya Sharma",
14    "effectiveDate": "2025-01-01",
15    "classificationId": 1,
16    "custId": 1
17  },
18  {
19    "custNameId": 5,
20    "customerNameType": "ALIAS",
```

PUT:

PUT

http://localhost:8080/api/customernames/6

Send

Docs

Params

Auth

Headers (8)

Body

Scripts

Settings

Cookies

raw

JSON

Schema

Beautify

```
1 {
2   "custNameId": 6,
3   "customerNameType": "PRIMARY",
4   "customerNameValue": "Dinesh kumar",
5   "effectiveDate": "2024-02-01",
6   "classificationId": 2,
7   "custId": 9
8 }
```

Body

200 OK

120 ms

305 B

Save Response

JSON

Preview

Visualize

```
1 {
2   "custNameId": 6,
3   "customerNameType": "PRIMARY",
4   "customerNameValue": "Dinesh kumar",
5   "effectiveDate": "2024-02-01",
6   "classificationId": 2,
7   "custId": 9
8 }
```

```
mysql> select * from customer_names;
```

customer_name_id	crud_value	customer_name_type	customer_name_value	effective_date	customer_classification_id	customer_identifier
2	C	PRIMARY	Rahul Menon	2024-04-01	3	2
3	D	PRIMARY	Sanjay Iyer	2024-01-01	2	3
4	C	PRIMARY	Priya Sharma	2025-01-01	1	1
5	C	ALIAS	Priya S. Sharma	2025-01-01	1	1
6	U	PRIMARY	Dinesh kumar	2024-02-01	3	2

5 rows in set (0.02 sec)

DELETE:

DELETE

http://localhost:8080/api/customernames/6

Send

Docs

Params

Auth

Headers (6)

Body

Scripts

Settings

Cookies

Body

200 OK

44 ms

184 B

Save Response

Raw

Preview

Visualize

```
1 Deleted successfully
```

```
mysql> select * from customer_names;
```

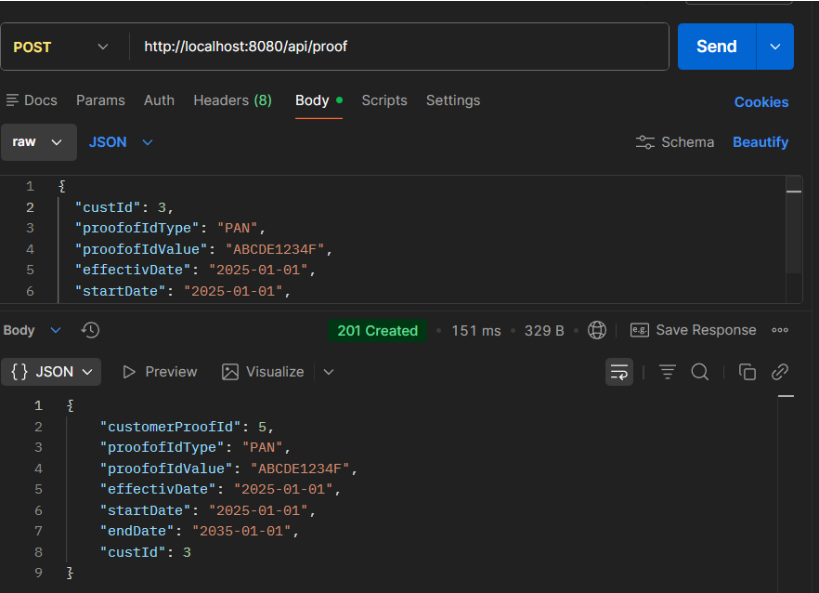
customer_name_id	crud_value	customer_name_type	customer_name_value	effective_date	customer_classification_id	customer_identifier
2	C	PRIMARY	Rahul Menon	2024-04-01	3	2
3	D	PRIMARY	Sanjay Iyer	2024-01-01	2	3
4	C	PRIMARY	Priya Sharma	2025-01-01	1	1
5	C	ALIAS	Priya S. Sharma	2025-01-01	1	1
6	D	PRIMARY	Dinesh kumar	2024-02-01	3	2

5 rows in set (0.01 sec)

Customer Proof of ID:

POST:

```
{
  "custId": 1,
  "proofofIdType": "PAN",
  "proofofIdValue": "ABCDE1234F",
  "effectivDate": "2025-01-01",
  "startDate": "2025-01-01",
  "endDate": "2035-01-01"
}
```

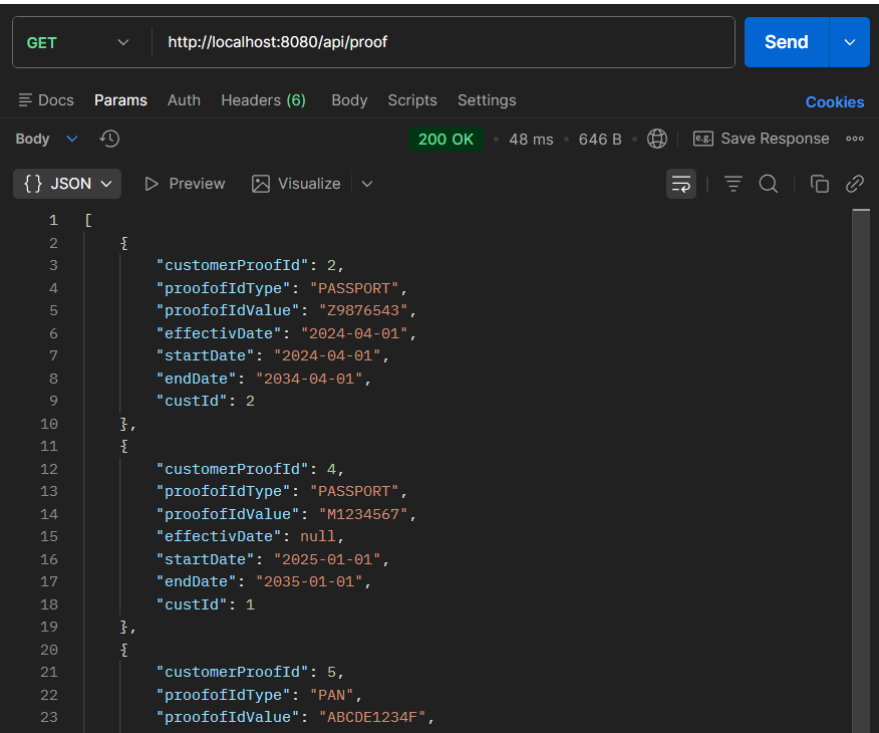


```
mysql> select * from customer_proof_ofId;
```

customer_proofofid_id	crud_value	effective_date	end_date	cust_proof_of_id_type	cust_proof_of_id_value	start_date	customer_identifier
2	C	2024-04-01	2034-04-01	PASSPORT	Z9876543	2024-04-01	2
3	D	2024-01-01	2034-01-01	PAN	SANPI1234K	2024-01-01	3
4	C	NULL	2035-01-01	PASSPORT	M1234567	2025-01-01	1
5	C	2025-01-01	2035-01-01	PAN	ABCDE1234F	2025-01-01	3

4 rows in set (0.01 sec)

GET:



PUT:

PUT

http://localhost:8080/api/proof/5

Send

Docs

Params

Auth

Headers (8)

Body

Scripts

Settings

Cookies

raw

JSON

Schema

Beautify

1

{

2

"customerProofId": 5,

3

"proofofIdType": "AADHAAR",

4

"proofofIdValue": "0000-8901-2345",

}

Body

200 OK

418 ms

332 B

Save Response

JSON

Preview

Visualize

1

{

2

"customerProofId": 5,

3

"proofofIdType": "AADHAAR",

4

"proofofIdValue": "0000-8901-2345",

5

"effectiveDate": "2024-01-10",

6

"startDate": "2024-01-10",

7

"endDate": "2034-01-10",

8

"custId": 3

9

}

mysql> select * from customer_proof_ofId;

customer_proofofid_id	crud_value	effective_date	end_date	cust_proof_of_id_type	cust_proof_of_id_value	start_date	customer_identifier
2	C	2024-04-01	2034-04-01	PASSPORT	Z9876543	2024-04-01	2
3	D	2024-01-01	2034-01-01	PAN	SANPI1234K	2024-01-01	3
4	C	NULL	2035-01-01	PASSPORT	M1234567	2025-01-01	1
5	C	2025-01-01	2035-01-01	PAN	ABCDE1234F	2025-01-01	3

4 rows in set (0.01 sec)

DELETE:

DELETE

http://localhost:8080/api/proof/5

Send

Docs

Params

Auth

Headers (6)

Body

Scripts

Settings

Cookies

Body

200 OK

229 ms

184 B

Save Response

Raw

Preview

Visualize

1

Successfully deleted

Cannot trigger the same request again as its already deleted

DELETE

http://localhost:8080/api/proof/5

Send

Docs

Params

Auth

Headers (6)

Body

Scripts

Settings

Cookies

Body

400 Bad Request

22 ms

167 B

Save Response

Raw

Preview

Debug with AI

1

Element already deleted

mysql> select * from customer_proof_ofId;

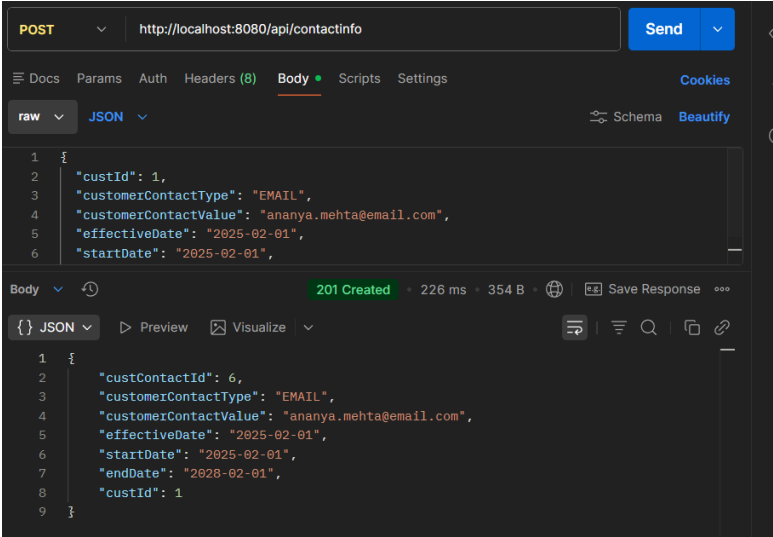
customer_proofofid_id	crud_value	effective_date	end_date	cust_proof_of_id_type	cust_proof_of_id_value	start_date	customer_identifier
2	C	2024-04-01	2034-04-01	PASSPORT	Z9876543	2024-04-01	2
3	D	2024-01-01	2034-01-01	PAN	SANPI1234K	2024-01-01	3
4	C	NULL	2035-01-01	PASSPORT	M1234567	2025-01-01	1
5	D	2024-01-10	2034-01-10	AADHAAR	0000-8901-2345	2024-01-10	3

4 rows in set (0.01 sec)

Customer contact information:

```
{
  "custId": 1,
  "customerContactType": "EMAIL",
  "customerContactValue": "ananya.mehta@email.com",
  "effectiveDate": "2025-02-01",
  "startDate": "2025-02-01",
  "endDate": "2028-02-01"
}
```

POST:



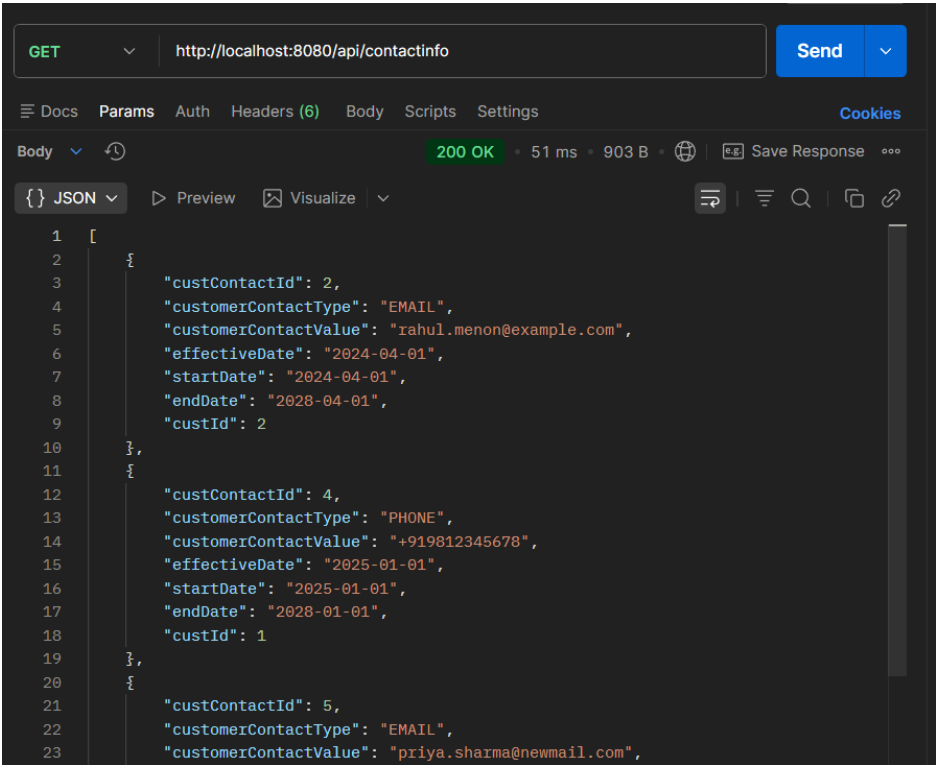
Database updated with the new post request

```
mysql> select * from customer_contact_info;
```

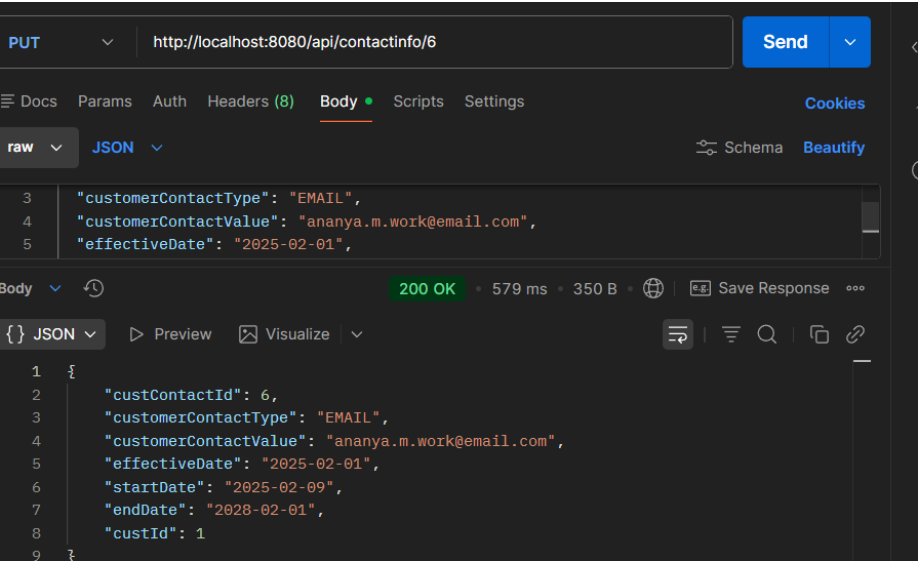
cust_contact_info_id	crud_value	customer_contact_type	customer_contact_value	effective_date	end_date	start_date	customer_identifier
2	C	EMAIL	rahul.menon@example.com	2024-04-01	2028-04-01	2024-04-01	2
3	D	EMAIL	sanjay.iyer@example.com	2024-01-01	2026-01-01	2024-01-01	3
4	C	PHONE	+919812345678	2025-01-01	2028-01-01	2025-01-01	1
5	C	EMAIL	priya.sharma@newmail.com	2025-01-01	2028-01-01	2025-01-01	1
6	C	EMAIL	ananya.mehta@email.com	2025-02-01	2028-02-01	2025-02-01	1

5 rows in set (0.01 sec)

GET:



PUT:



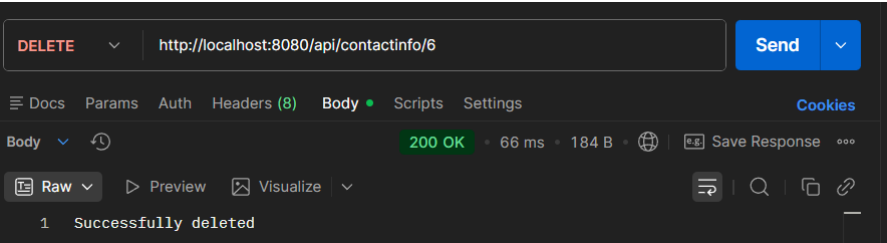
On changing the email and sending put request the database is updated

```
mysql> select * from customer_contact_info;
```

cust_contact_infoid	crud_value	customer_contact_type	customer_contact_value	effective_date	end_date	start_date	customer_identifier
2	C	EMAIL	rahul.menon@example.com	2024-04-01	2028-04-01	2024-04-01	2
3	D	EMAIL	sanjay.iyer@example.com	2024-01-01	2026-01-01	2024-01-01	3
4	C	PHONE	+919812345678	2025-01-01	2028-01-01	2025-01-01	1
5	C	EMAIL	priya.sharma@newmail.com	2025-01-01	2028-01-01	2025-01-01	1
6	U	EMAIL	ananya.m.work@email.com	2025-02-01	2028-02-01	2025-02-09	1

5 rows in set (0.01 sec)

DELETE:



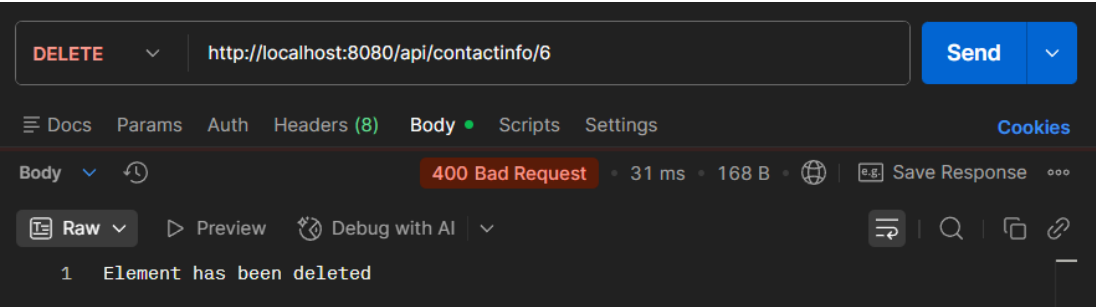
Database is updated with the new CRUD value

```
mysql> select * from customer_contact_info;
```

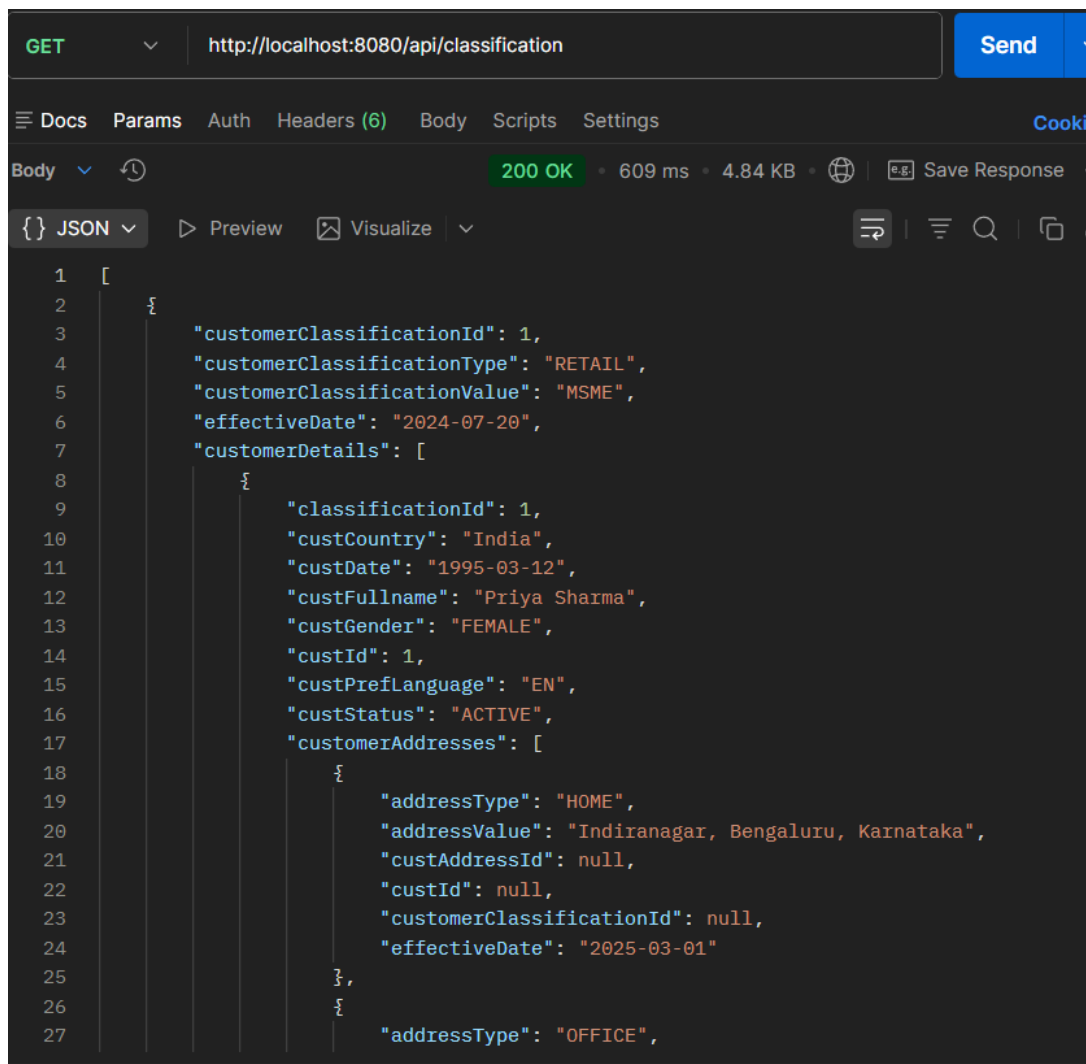
cust_contact_infoid	crud_value	customer_contact_type	customer_contact_value	effective_date	end_date	start_date	customer_identifier
2	C	EMAIL	rahul.menon@example.com	2024-04-01	2028-04-01	2024-04-01	2
3	D	EMAIL	sanjay.iyer@example.com	2024-01-01	2026-01-01	2024-01-01	3
4	C	PHONE	+919812345678	2025-01-01	2028-01-01	2025-01-01	1
5	C	EMAIL	priya.sharma@newmail.com	2025-01-01	2028-01-01	2025-01-01	1
6	D	EMAIL	ananya.m.work@email.com	2025-02-01	2028-02-01	2025-02-09	1

5 rows in set (0.00 sec)

Cannot send the delete request again



Customer classification is now updated with the details and is no more NULL



The screenshot shows a REST client interface with a GET request to `http://localhost:8080/api/classification`. The response is a 200 OK status with a response time of 609 ms and a body size of 4.84 KB. The response body is a JSON object representing a customer classification and their details.

```
1  [
2    {
3      "customerClassificationId": 1,
4      "customerClassificationType": "RETAIL",
5      "customerClassificationValue": "MSME",
6      "effectiveDate": "2024-07-20",
7      "customerDetails": [
8        {
9          "classificationId": 1,
10         "custCountry": "India",
11         "custDate": "1995-03-12",
12         "custFullname": "Priya Sharma",
13         "custGender": "FEMALE",
14         "custId": 1,
15         "custPrefLanguage": "EN",
16         "custStatus": "ACTIVE",
17         "customerAddresses": [
18           {
19             "addressType": "HOME",
20             "addressValue": "Indiranagar, Bengaluru, Karnataka",
21             "custAddressId": null,
22             "custId": null,
23             "customerClassificationId": null,
24             "effectiveDate": "2025-03-01"
25           },
26           {
27             "addressType": "OFFICE",
```

In this lab we have demonstrated the entire backend code using the layered approach which enable loose coupling and enables expansion.

All the APIs created have been demonstrated above.